Elicited Confidence in Clinical Language Models Encodes Evidence Sufficiency, Not Answer Correctness

Anonymous

Abstract—Confidence-gated deferral—routing low-confidence outputs to a clinician—is the most widely proposed safety mechanism for clinical language models, and it presumes that a model's stated confidence reflects whether its answer is right. We show it reflects something else: whether the model has the information it needs. In a causal manipulation replicated across three models and two clinical task types, we remove a single item a case requires—a key diagnostic finding, or a laboratory value without which a calculation cannot be performed—and restore it. Removal moves accuracy by 9 points on diagnostic vignettes and 41 on calculations, and moves stated confidence by up to 3.6 points, so the model registers the deficiency. Its ability to separate correct answers from incorrect ones does not move detectably in any of the five cells, with every interval bounded inside $\pm.17$ AUROC. Grading the manipulation separates the two responses further: removing one, two and three required inputs lowers confidence monotonically, within item for 95% of items, and lowers accuracy from 0.73 to 0.25, while discrimination is unchanged across the three levels ($+.014$ $[-.089, +.112]$). The consequence is that a confidence gate catches information gaps and not knowledge errors. Across ~34,200 episodes, five models and seven evaluation sets, discrimination is strong where insufficiency dominates—interactive consultation, 0.75–0.80—and at chance where the information is complete and the error is one of knowledge (0.48–0.54). A deferral threshold meeting an 0.85 service level in validation returns 0.35–0.50 at the capability frontier while still deferring up to four fifths of cases, and nothing in the validation data signals the shortfall.

Index Terms—Clinical decision support, large language models, uncertainty quantification, selective prediction, patient safety, model calibration.

I. Introduction

LARGE language models are entering clinical decision support through triage assistants, differential-diagnosis aids, and documentation tools. Because such systems will be wrong some of the time, essentially every proposed deployment architecture includes an abstention pathway: the model reports a confidence, and outputs below a threshold are escalated to a clinician. This design is attractive because it requires no access to model internals—the confidence is simply elicited in natural language—and because it maps onto existing clinical triage workflows.

A. The untested premise

Confidence-gated deferral is only a safety mechanism if the confidence it gates on is informative about correctness. In the general machine-learning literature this premise has been repeatedly questioned: verbalized confidence is systematically overconfident [3], [2], its structure across models is dominated by a single shared item-difficulty factor rather than by model-specific self-knowledge [14], and overconfidence has been traced to a failure to condition the stated confidence on the model's own answer [6]. That literature evaluates aggregate calibration on benchmark question sets. It cannot address the question a clinical deployment actually turns on, because it lacks clinical risk labels: does confidence remain informative on the cases where being wrong is most consequential?

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B. Why the clinical case is different

A safety mechanism that works on average but fails on the hardest cases is worse than no mechanism, because it produces false assurance exactly where assurance is unwarranted. Whether that describes confidence gating cannot be settled by measuring calibration more carefully; it requires knowing what the confidence is a report of. Two features of the clinical setting make that question answerable here and not on general benchmarks.

First, a clinical case has a determinate information requirement: a diagnosis turns on a specific finding, a dose on a specific laboratory value. One can therefore remove exactly what the case needs and observe what moves—an intervention rather than a correlation. General-knowledge benchmarks do not admit it, because there is no defined item whose removal makes the question unanswerable rather than merely different.

Second, clinical deployment spans two regimes that make opposite demands on such a signal. A consultation agent does not receive a fully specified vignette; it decides what history to take and which tests to order, and is therefore responsible for the evidence its own confidence rests on. A system reading a completed record is not, and its residual errors are of a different kind. Whether a gate validated in one regime holds in the other is an empirical question that has not been asked, and Section V-N shows it has a price.

C. Scope of this study

The study is organised around one intervention and its consequences. We manipulate the information a case supplies—withheld, supplied, or obtainable only by the model's own action—on two clinical task types, and grade the manipulation by amount removed. We then evaluate the resulting account against three deployment regimes that differ in what makes the model wrong: the capability frontier, where the vignette is complete and the error is one of knowledge; complete vignettes and procedural calculations at standard difficulty; and interactive consultation, where the model must obtain its own evidence. The elicitation protocol is byte-identical across all of them, so a difference in discrimination is attributable to the regime rather than to how the question was asked, and we control for the artifacts that could produce the result spuriously—class balance, response format, elicitation position, response scale, and the scoring rule for free-text diagnoses.

D. Measurement choice

Calibration in clinical machine learning is usually summarised by expected calibration error, and a substantial literature reports ECE for medical language models. ECE is the wrong instrument for the question posed here. A model that subtracts a constant from every stated confidence improves its ECE without becoming any better at identifying which of its answers are wrong, and it is the latter property that a deferral gate consumes. The two can move in opposite directions: in our data the second-vendor model has the better ECE at the capability frontier while having the worse discrimination, and its apparently graceful degradation turns out to be a uniform downward shift carrying no item-level information. We therefore report discrimination as the primary endpoint, give ECE alongside it, and make the deployment consequence explicit through decision curves.

E. Contributions

- We identify what elicited confidence in clinical models reports, by removing and restoring the single item a case requires and observing what moves. Stated confidence tracks the removal; its ability to discriminate correct from incorrect answers does not shift detectably, and is bounded inside $\pm.17$ AUROC in every cell. The result holds for a diagnostic finding and for a computational input, on three models, and is the paper's central claim. Grading the manipulation sharpens it: removing one, two and three required inputs lowers stated confidence monotonically—within item, for 95% of items—while leaving discrimination unchanged across the three levels.
- We show this account predicts where confidence works and where it fails, rather than merely describing it: discrimination is strong in interactive consultation, where the model may simply not have asked, and at chance at the capability frontier, where the information is present and the error is one of knowledge. Within a single model the range is 0.48 to 0.80—wider than the range across models within any one condition.
- We quantify the deployment consequence. A deferral threshold calibrated to an 0.85 service level delivers 0.35–0.50 at the capability frontier, from every calibration origin tested, while deferring between a fifth and four fifths of cases; no threshold reaches the service level there at all. The shortfall is invisible in the validation data, and the direction of transfer a development pipeline naturally takes—validate where the model is competent, deploy where it is not—is the failing one.

II. Related Work

A. Eliciting and calibrating confidence in language models

Kadavath et al. [2] established that models can, under favorable conditions, self-evaluate proposed answers. Black-box elicitation was then systematized by Xiong et al. [3], who factored the problem into prompting strategy, sampling, and aggregation and benchmarked the combinations across five datasets; they report that every method examined degrades on tasks demanding professional knowledge—the regime this paper isolates. Subsequent work has largely pursued mitigation. Self-evaluation can be adapted for selective prediction [4]; calibration can be improved by self-generated distractors [7], by aligning verbalized with internal confidence [8], [10], by order-aware alignment [9], or by self-calibration at test time [5]. ADVICE [6] identifies answer-independence—the failure to condition the stated confidence on the model's own answer—as a primary driver of overconfidence and corrects it by fine-tuning. Surveys and empirical audits confirm that overconfidence persists across current models [11], [12], [21].

B. Is elicited confidence metacognition?

A separate line asks what elicited confidence measures rather than how to fix it. MIRROR [13] benchmarks metacognitive calibration hierarchically; other work probes introspection directly [15] or attempts to steer it [16]. Most relevant to us, Moran et al. [14] decompose binary confidence judgements from 20 models across six benchmarks and find the cross-model confidence matrix approximately rank-one: models share an item-level difficulty axis and differ mainly in decision threshold, with the confidence–performance relationship vanishing once items all models agree on are removed. They report no evidence of individuated metacognition in any domain tested. A complementary result shows that confidences do not govern behavior: models act against their own stated confidences in agentic settings [17]. Our general-domain observations—strong cross-model agreement in confidence, near-zero within-model correlation with correctness—reproduce this picture, and we treat it as established background rather than as a contribution. What this line cannot address, lacking clinical risk labels, is whether confidence remains informative where error is most consequential.

C. Behavioral alternatives to verbalized confidence

Because stated confidence is unreliable, many systems substitute behavioral signals: answer consistency across samples [18], conformal procedures built on self-consistency sampling [19], or reasoning-conditioned estimates [20]. Related lines predict reasoning length ahead of time [40], distil black-box uncertainty [41], aggregate evidence in a Dirichlet framework [42], make elicitation noise-aware for retrieval-augmented settings [43], and evaluate calibration by semantic sampling [44]. Selective prediction on these signals has been studied for code [45], [22], for deferral at the edge [39], and for vision-language triage in the medical setting [33]. We evaluate the two cheapest behavioral alternatives—sampling self-consistency and cross-model agreement—and find neither recovers discrimination at the capability frontier.

D. Test-time compute and its effect on confidence

Medical test-time scaling has been studied on the ascending half of the compute curve [25], [26]. Two results bear directly on our design. CDUR [20] reports that increasing chain-of-thought budget can induce overconfidence non-monotonically, and Hu et al. [27] show that enabling thinking shifts the composition of errors while aggregate accuracy barely moves. We therefore vary the effort parameter as a manipulation and verify that our conclusions hold at every operating point rather than at one.

E. Uncertainty and safety in clinical language models

Confidence in clinical LLMs has been examined by specialty [28] and by prompting strategy [29], generally reporting aggregate calibration on benchmark question sets. SaFE-Scale [30] shows that clinical safety and accuracy follow different scaling laws across deployment conditions including inference-time compute, and that consequential errors concentrate in a small subset of items—motivating our stratification by clinical risk rather than by average performance. Failure-mode analyses of clinical agents identify information-seeking failure, anchoring, and premature closure [31], [35], [34], and ResidencyRL [32] establishes missed-red-flag rate as a headline safety endpoint. That line of work measures how often clinical models fail; we ask whether the models' own stated confidence anticipates it. Reasoning-model behaviour on medical tasks has been characterised directly [46], and agentic scaffolding studied for diagnostic instruction [47], [48]. Multi-agent deliberation has been proposed to improve clinical reasoning [37], [36], and long-context clinical benchmarks expose anchoring to early interpretations [38]. Interactive clinical benchmarks [1] provide the consultation modality but have not been used to evaluate confidence, which is the gap our interactive experiments fill.

III. Materials and Methods

A. Models

Five models from two vendors spanning three capability tiers were evaluated: gpt-5.4, gpt-5.4-mini, gpt-5.4-nano, gpt-4o-mini (non-reasoning), and claude-sonnet-5. Including a non-reasoning model tests whether the phenomenon depends on explicit reasoning machinery. All calls were made through vendor APIs between 2026-08-28 and 2026-08-29; every response, token count, and latency was logged and cached.

B. Datasets

Clinical. MedAgentsBench [23] provides a hard subset (test_hard, 894 items over 10 sources) on which contemporary models sit near chance, and a standard-difficulty split from the same sources; using both isolates difficulty from corpus. AgentClinic [1] provides 214 OSCE vignettes with free-text gold diagnoses, used both as static vignettes and as interactive consultations. MedCalc-Bench [24] provides 1,100 procedural calculation items with an acceptance band and a ground-truth formula, permitting separation of execution slips from formula errors.

Non-clinical controls. MMLU-Pro and ARC-Challenge were included as independent corpora spanning a wide accuracy range.

C. Threats addressed by design

Four alternative explanations of a null discrimination result are addressed by construction rather than by argument. Class balance: handled by (3). Response format: handled by the four-format control and by the free-response corpora. Poor verbalisation of an internal signal: handled by comparing against sampling self-consistency, which requires no verbalisation at all. Censoring by response ceilings: handled by the escalation rule in Section IV-A, which matters because a ceiling set near the reasoning draw produces non-responses concentrated at high effort and would invert the direction of any effort-related result. A fifth threat, that a condition differs in difficulty rather than in kind, is the reason every cross-condition contrast is made against accuracy-matched controls (Section IV-D) rather than against raw sets.

TABLE I
Models evaluated. Vendor, scale and architecture are varied so that a condition effect cannot be attributed to any one of them. Operating points are those surviving the separation test of Section IV-B.

Model	Vendor	Tier	Reasoning	Operating points
gpt-5.4	OpenAI	large	yes	xhigh, high, medium, low, none
gpt-5.4-mini	OpenAI	mid	yes	xhigh, high, low, none
gpt-5.4-nano	OpenAI	small	yes	xhigh, high, low, none
gpt-4o-mini	OpenAI	mid	no	instructed-token protocol
claude-sonnet-5	Anthropic	large	yes	max, xhigh, high, medium, low

D. Model selection rationale

Table I lists them. The five models are chosen to vary three factors that could each explain a confidence result on their own. Vendor is varied because tokenisation, instruction tuning, and refusal behaviour are shared within a vendor and would make a single-vendor finding uninformative about the phenomenon. Scale is varied

TABLE II
Interactive versus static presentation of the same 214 AgentClinic cases, gpt-5.4-mini. Evidence acquisition changes what the confidence signal is worth; the case material, gold labels and elicitation schema are identical.

Presentation	n	Acc.	AUROC	[95% CI]
Static vignette	409	.655	.669	[.617, .721]
Interactive, $T = 8$	642	.632	.751	[.713, .787]
gpt-5.4				
Static vignette	303	.667	.689	[.628, .748]
Interactive, $T = 8$	642	.727	.800	[.759, .840]

across three tiers within one family, so that a result cannot be attributed to a particular capability level. Architecture is varied by including a model without explicit reasoning machinery: if elicited confidence were a readout of an internal deliberation process, the phenomenon should differ between models that perform such deliberation and models that do not. All three factors are crossed with the difficulty manipulation, which is what allows condition effects to be separated from model effects.

E. Elicitation protocol

Every episode used a fixed structured-output schema requesting, in one response: the differential actually considered, the findings used, the committed answer, and a confidence. The schema is byte-identical across all conditions, so the elicitation of confidence never varies with the experimental factor. Responses were parsed from JSON with at most one retry; parse rate was 1.000.

Three properties of this design matter for interpretation. First, the schema requests the differential before the answer and the confidence, so the model states what it considered before committing and the differential is not a post-hoc rationalisation of a chosen answer. Second, nothing in the schema references difficulty or the experimental condition, so the elicitation cannot leak the manipulation. Third, because the schema is fixed, the elicited differential is obtained under identical instructions everywhere; a condition that produced tersely worded answers cannot thereby appear to consider fewer hypotheses, which is the artifact that would otherwise confound any breadth-based measure.

Elicitation-format control. To exclude the possibility that null discrimination reflects a poorly chosen response format, four formats were compared on identical items: an integer 1–10 scale, a 0–100 probability, a four-level verbal scale, and a post-hoc self-check in which the answer and the confidence judgement are produced in separate calls.

F. Interactive protocol

In the interactive setting the model receives only the presenting information and must issue at most $T = 8$ actions, each either a question to a simulated patient or a test order, before committing. Patient responses are produced by a separate model; test results are returned by deterministic lookup against the case file, so evidence availability cannot drift between conditions. The commitment turn uses the same fixed schema as all static conditions. The response ceiling was set well above the observed reasoning draw; truncation was logged per episode and any truncated call was automatically retried at a doubled ceiling, because a ceiling near the reasoning draw manufactures non-responses at high effort.

G. Formal setup

Let $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^{N}$ be a set of clinical items, x_i the presentation and y_i the reference answer. A model m under operating point e produces the episode of (1),

$$z_i^{(m,e)} = \big(\hat{y}_i,\ \mathcal{H}_i,\ c_i\big) \sim \pi_{m,e}(\cdot \mid x_i), \tag{1}$$

where $\hat{y}_i$ is the committed answer, $\mathcal{H}_i$ the elicited differential, and $c_i \in \{1, \ldots, 10\}$ the elicited confidence. Correctness is $a_i = \mathbb{1}[\hat{y}_i = y_i]$. All fields are emitted in a single structured response whose schema is invariant across every condition studied, so $\pi_{m,e}$ differs between conditions only in e and in $\mathcal{D}$.

H. Discrimination

The primary endpoint is the probability that a correct episode is assigned higher confidence than an incorrect one,

$$\mathrm{AUC}(c, a) = \Pr\left(c_i > c_j\right) + \tfrac{1}{2}\Pr\left(c_i = c_j\right) \tag{2}$$

for $a_i = 1$, $a_j = 0$, estimated by the rank statistic $\widehat{\mathrm{AUC}} = \big(\sum_{i:a_i=1} r_i - n_1(n_1+1)/2\big)/(n_1 n_0)$ with r_i the mid-rank of c_i in the pooled sample and n_1, n_0 the counts of correct and incorrect episodes.

Reliability diagrams for each condition are given in Fig. 6 and risk–coverage curves in Fig. 7. We report (2) rather than expected calibration error $\mathrm{ECE} = \sum_b \frac{|B_b|}{n}\left|\bar{a}_{B_b} - \bar{c}_{B_b}\right|$ because ECE conflates two failures that have opposite deployment consequences. A model that shifts every confidence downward by a constant improves ECE without gaining any ability to identify its own errors, since (2) is invariant to any strictly monotone reparameterisation of c. Both are reported; only (2) bears on deferral.

I. Base-rate control

Because accuracy differs across conditions, a difference in (2) could in principle reflect class balance. Although the rank estimator is base-rate invariant in population, we verify this empirically. Given a target set with accuracy α^* and a comparison set $\mathcal{C} = \mathcal{C}^+ \cup \mathcal{C}^-$, we retain all of $\mathcal{C}^-$ and draw

$$n^+ = \left\lfloor |\mathcal{C}^-| \cdot \frac{\alpha^*}{1 - \alpha^*} \right\rfloor \tag{3}$$

correct episodes without replacement, repeating $B_{\mathrm{m}} = 3000$ times and reporting the distribution of $\widehat{\mathrm{AUC}}$ over the matched resamples.

J. Scoring free-text diagnoses

AgentClinic answers are free text, so equality with the reference is not the right test. Writing $A(y_i)$ for the reference diagnosis and its alias set,

$$a_i = \mathbb{1}\big[\exists\, v \in A(y_i):\ \mathrm{sim}(\hat{y}_i, v)\big], \tag{4}$$

where $\mathrm{sim}(u, v)$ holds when at least half the content tokens of v, after stop-word removal and light stemming, appear in u. Alias sets are fixed before any model is run. The same rule scores the elicited differential $\mathcal{H}_i$ when we ask whether the reference diagnosis was entertained at all.

K. Deferral utility

A gate with threshold τ routes episodes with $c_i < \tau$ to a clinician. Writing $R(\tau) = \{i : c_i \geq \tau\}$ for the retained set, we report the deferral fraction $d(\tau) = 1 - |R(\tau)|/N$ against the endpoint rate on retained cases,

$$q(\tau) = \frac{1}{|R(\tau)|} \sum_{i \in R(\tau)} a_i, \tag{5}$$

and the gain $q(\tau) - q(0)$. A gate has clinical value only if $q(\tau)$ rises appreciably for $d(\tau)$ small enough to be absorbed by the workflow; $q(\tau) \leq q(0)$ at any operating τ means the gate is not merely useless but harmful, since it consumes clinician time while leaving the retained set no safer.

L. Answer stability

To characterise what c_i tracks we define, for model m over the set E of operating points,

$$s_i^{(m)} = \frac{1}{|E|} \max_{v}\ \big|\{e \in E :\ \hat{y}_i^{(m,e)} = v\}\big|, \tag{6}$$

the modal-answer share across operating points. The circularity concern—that $s^{(m)}$ and $c^{(m)}$ are computed from the same episodes—is addressed by regressing $c^{(m)}$ on $s^{(m')}$ for $m' \neq m$, which shares no episodes with the outcome.

M. Non-response

An episode without a parseable commitment is not an error, and scoring it as one would confound answer quality with output-format compliance. Non-responses are recorded as a separate quantity NR_i and reported per condition; they are zero in every condition analysed here, which matters because the two mechanisms that generate them—response-ceiling truncation and refusal— are both correlated with the experimental factor and would bias the endpoint if silently dropped or silently counted as wrong.

N. Uncertainty quantification

All intervals are percentile bootstraps over $B = 4000$ resamples. The resampling unit is the episode for endpoints computed over episodes and the item elsewhere, since episodes of one item under different operating points are not independent; contrasts between conditions evaluated on the same items resample items, so the interval reflects the repeated-measures design. Difference claims are tested by Wilcoxon signed-rank where paired and by permutation on the condition label where not. Because a confidence interval containing zero is not evidence of no effect, claims of no difference are tested by two one-sided tests against an equivalence margin of 0.10 AUROC, fixed in advance. Difference and equivalence tests are Holm-corrected as separate families (Section V-M).

IV. Experiment Configurations

A. Implementation details

All models were accessed through vendor APIs between 2026-08-28 and 2026-08-29. Reasoning-class models expose an ordinal effort parameter; the non-reasoning model does not, and is driven by an instructed-token protocol instead. Sampling temperature is left at the vendor default for reasoning models, which reject explicit temperature, and fixed at 0.3 for the non-reasoning model. Every request and response, with token counts and latency, is written to a content-addressed disk cache keyed on model, prompt, system message, operating point, seed, response ceiling, and a fingerprint of the output schema; the schema fingerprint is required because conditions that differ only in schema would otherwise collide on one cache entry.

Response ceilings are set well above the observed reasoning draw for each operating point. A ceiling near that draw is a censoring instrument rather than a measurement: at a 20,000-token ceiling the top operating point returned empty answers on 12.7% of episodes, exactly matching its truncation rate, which would manufacture a non-response effect at high effort. Any truncated call is automatically retried once at a doubled ceiling before truncation is accepted, and residual truncation is logged per episode; it is zero in all reported conditions. Rate-limit responses are retried on a separate budget from error retries, because sharing one budget causes attrition concentrated in the most token-hungry cells—measured at 23 lost episodes at the top operating point against 6 at the bottom before this was separated.

B. Operating points

Provider effort levels are nominal settings, not token budgets, and adjacent levels are not guaranteed to differ. Before use, each model's ladder is pruned by a separation test requiring both Cliff's $\delta \geq 0.33$ and a Holm-corrected rank-sum test between adjacent levels on realised reasoning tokens. On gpt-5.4-mini the medium level was indistinguishable from high ($\delta = 0.175$, $p_{\mathrm{Holm}} = 0.28$) and was merged; on claude-sonnet-5 the thinking-disabled mode was excluded from the ladder entirely, because disabling thinking raises visible output (median 460 versus 99 tokens)—the model relocates reasoning into the answer rather than eliminating it, so the mode is not the bottom of the same scale.

C. Datasets and splits

Table III summarises the evaluation sets. Difficulty is manipulated within corpus by using the hard and standard splits of the same sources, so difficulty is not confounded with domain.

TABLE III
Evaluation sets. Difficulty is varied within corpus (hard versus standard split of the same sources) so that difficulty is not confounded with domain. Episodes are the analysed units; items appear at several operating points and seeds.

Set	Domain	Items	Episodes	Answer format
MedAgentsBench-hard	clinical	894	8195	option letter
MedAgentsBench-std	clinical	400	3198	option letter
AgentClinic (static)	clinical OSCE	214	712	free-text diagnosis
AgentClinic (interactive)	clinical OSCE	214	1284	free-text diagnosis
MedCalc	clinical procedural	300	581	numeric
MMLU-Pro	general knowledge	300	600	option letter
ARC-Challenge	science	300	600	option letter

D. Cost and compute

The study uses inference only; no model was trained or fine-tuned. Total API expenditure was $156 across three days, dominated by the top operating point of the strongest model, which draws a median of 8,541 reasoning tokens per episode against 0 at the bottom of the ladder. Throughput rather than cost was the binding constraint: the principal model is rate-limited to 200,000 tokens per minute on the account used, giving roughly 25 episodes per minute at the top operating point, so grids were scheduled in token-minutes rather than call counts. Running two vendors concurrently, which draw on independent rate limits, raised sustained throughput to about 68 episodes per minute. All results are reproducible from the cache without further expenditure.

E. Evaluation criteria

Answers are extracted from the structured response; parse rate is 1.000 across all reported conditions. Multiple-choice items are scored by option identity, free-response numeric items against the benchmark's own acceptance band, and free-text diagnoses by the alias-aware match of (4). Episodes without a parseable commitment are recorded as non-responses and are reported separately rather than being scored as errors.

V. Results

The central experiment is presented first. Section V-A removes and restores the single item a case requires and reports what moves; Section V-B shows three episode pairs verbatim; Section V-C grades the manipulation, removing one, two and three required inputs; and Section V-D states what the surviving signal is. Sections V-E–F show that this account predicts where confidence works and where it fails. Sections V-G–L remove the artifacts that could produce those observations spuriously— including the possibility that the failure is one of verbalization rather than of information (Section V-H)—Section V-M reports the significance and equivalence tests, and Section V-N gives the deployment consequence.

A. Removing what a case requires moves confidence but not discrimination

The central experiment asks what elicited confidence is a report of, by removing the single item a case requires and observing which quantities move. The case, the reference label, the prompt and the elicitation schema are held fixed; only what the model has in hand varies. In the diagnostic setting the key discriminating finding is removed from the vignette, restored to it, or made available only if the model requests it. In the computational setting a numerically required laboratory value is removed, which makes the calculation unanswerable rather than merely harder.

Fig. 1 gives the result in full. The manipulation is effective in every cell and its magnitude is graded by how much the removal costs. On MedCalc, where the removed value makes the calculation unanswerable, accuracy falls by 41 points and mean stated confidence by 3.5 points on a ten-point scale. On AgentClinic, where the case remains answerable without the key finding, accuracy falls by 9 points and confidence by 0.5. The model therefore registers not merely that something is absent but roughly how much its absence matters, and it does so consistently across all three models tested, which span two orders of magnitude of scale.

Discrimination does not follow. Every interval contains zero (+.006, −.059, +.054, +.029, +.062), and the one that comes closest to excluding it—the MedCalc contrast on gpt-5.4-mini, +.062 $[-.000, +.127]$—has a lower bound that changes sign with the resampling seed, so we do not read it as a positive result. The bound rather than the point estimate is the finding: restoring information worth 41 points of accuracy shifts discrimination by no more than .17 in either direction across all five cells, and by no more than .13 in the direction that would favour it. Section V-M reports this as an equivalence test and states what it does and does not establish. A signal that moves by 3.5 points when the required input is removed, while its ability to separate right answers from wrong ones does not move, is reporting the availability of the input and not the correctness of the conclusion drawn from it (Table X).

One contrast does move discrimination. Requiring the model to obtain the finding through its own action, rather than receiving it, adds +.107 $[+.008, +.204]$ on gpt-5.4 and non-significant increments of +.039 and +.049 on the two smaller models—the same sign throughout, reaching significance only where the model is strong enough for its own actions to be informative. Evidence agency, unlike evidence availability, leaves a trace in the confidence signal—consistent with the interpretation below, since a model that had to ask has direct evidence about what it does and does not hold.

Two further observations locate what the signal is. First, within the acquired condition, confidence distinguishes episodes in which the model actually obtained the key finding from those in which it did not at AUC = 0.661: the model has partial access to whether its own information is sufficient. Second, discrimination is higher among episodes

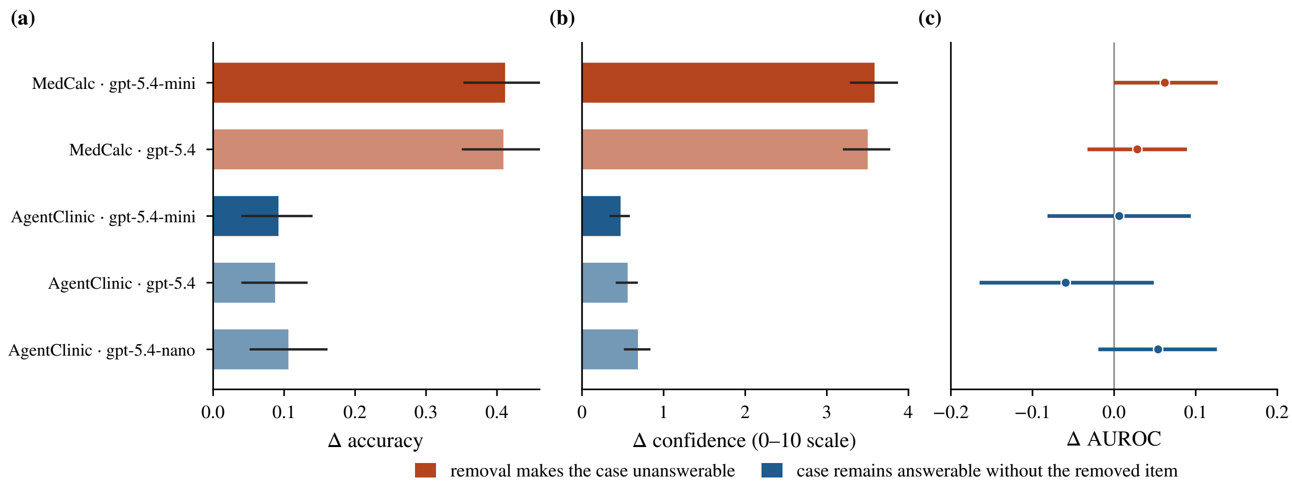

Fig. 1. The central result. Restoring the single item a case requires moves accuracy (a) and stated confidence (b) by amounts graded to how much the removal cost—large where the case was rendered unanswerable, small where it remained answerable—while leaving discrimination statistically unmoved (c; bars are 95% bootstrap intervals on ΔAUROC, all five containing zero). Elicited confidence tracks what the model is missing, not whether its answer is right.

where the key finding was not obtained (0.790) than where it was (0.714), even though accuracy is much lower there (0.543 versus 0.727).

Taken together these say that elicited confidence reports an information state rather than an answer state. It tracks, with reasonable fidelity, whether the model has what it needs; it does not track whether the conclusion it drew is correct. The two coincide only when insufficiency is the dominant source of error—which is the case in interactive consultation, where the model may simply not have asked, and is not the case at the capability frontier, where the information is fully present and the error is one of knowledge.

B. Three episodes, verbatim

The quantities above are averages over conditions; Fig. 2 shows what a single episode pair looks like, chosen to illustrate the three regimes the account predicts.

In the first, removal is the reason the model is wrong, and its confidence says so: the barium enema is what distinguishes Hirschsprung disease from functional constipation, and without it the model commits to constipation at 7/10 while listing the reference diagnosis second. Restored, it commits correctly at 10/10.

The second is the case that a confidence gate cannot catch. The model is wrong in both conditions, and states 9/10 in both. Restoring the platelet count does bring thrombotic thrombocytopenic purpura into the elicited differential—the information was received and used—but the model still commits to a tropical infection. The error is one of weighting evidence already in hand, and nothing in the stated confidence distinguishes this episode from the one above.

The third is the paper's claim in a single case. Anoscopy findings move the confidence from 8 to 10 while the committed diagnosis is correct either way. The two points of confidence buy no correctness because there was none to buy; they report that the model now has what the case calls for.

C. Confidence responds to how much is missing, discrimination does not

The manipulation above is binary. If elicited confidence reports an information state, the response should be graded, so we removed 1, 2 and 3 of the required inputs from the same MedCalc note. Restricting to the 85 items carrying at least three removable inputs makes every dose level a within-item comparison.

Stated confidence declines monotonically with the number of inputs removed—9.29, 7.47, 6.66, 5.33—and the decline is a within-item effect rather than a shift of the group mean: taking each item from a complete note to three removals, confidence falls for 160 of 169 item–seed pairs, by 3.98 points on average (Spearman $\rho = -0.582$, $p < 10^{-60}$; Fig. 3a). Accuracy falls alongside it, 0.728 to 0.253.

Discrimination does not follow. Across the three removal levels it is 0.654, 0.573 and 0.669; the contrast between removing one input and removing three is +.014 $[-.089, +.112]$, while over the same interval confidence falls 2.14 points and accuracy falls 16 points. Doubling and tripling the deficiency is registered in full by the confidence and not at all by its ability to sort right answers from wrong ones.

One step does move discrimination: going from a complete note to a single removal costs −.175 $[-.269, -.076]$ on this subset. These are the most input-dense calculations in MedCalc, and the effect is specific to them—the same contrast over the full set of items carrying at least one removable input is +.063, with an interval touching zero (Table X). Where the model is asked to compute from many values at once, losing any of them does cost some discrimination; losing more of them costs none.

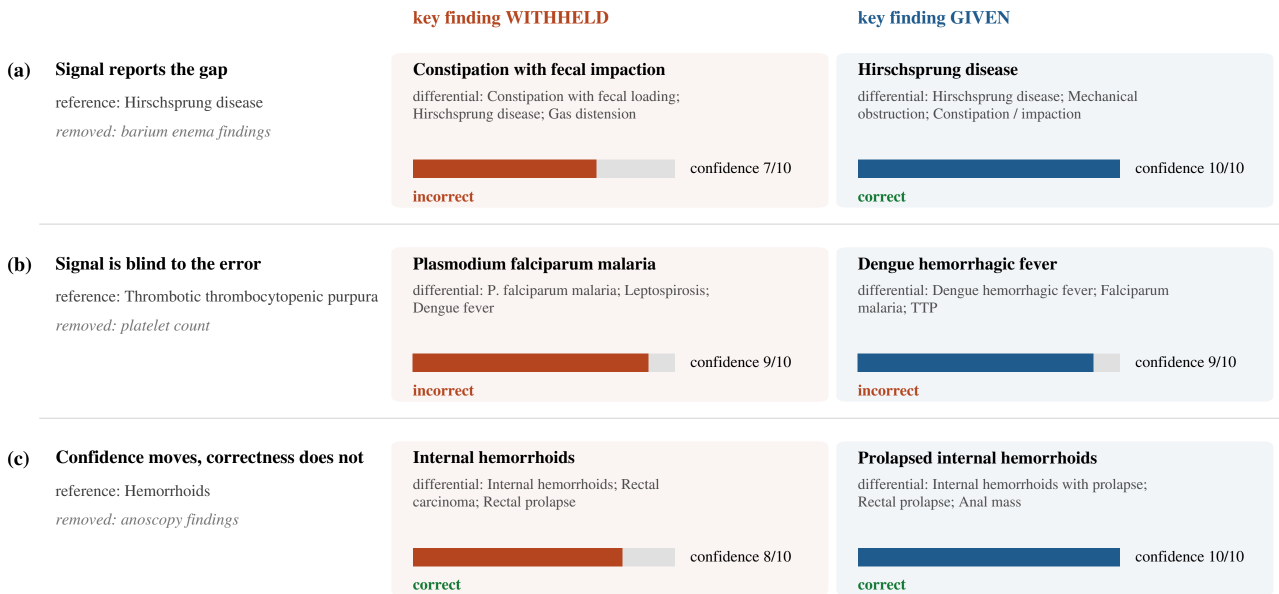

Fig. 2. Three episode pairs from the removal manipulation, verbatim, gpt-5.4-mini. Each row is one case under the two conditions; the committed diagnosis, the top of the elicited differential, and the stated confidence are shown as recorded. (a) The removal creates a gap the signal reports: without the barium enema the model commits to constipation at 7/10, with it to the reference diagnosis at 10/10. (b) The removal is not what makes the model wrong: restoring the platelet count brings the reference diagnosis into the differential but the model still commits elsewhere, at the same 9/10 both times. (c) The pattern the paper is about: the confidence moves by two points while the answer stays correct in both conditions—the signal is reporting the information state, not the answer.

D. What elicited confidence tracks

To characterise the signal that survives the manipulation, we examine how elicited confidence relates to three quantities computed independently of it: another model's confidence on the same item, the model's own correctness, and the answer stability of (6) computed from a different model. Fig. 8 summarises the three relationships. Confidence agrees across models more strongly than accuracy does: mean pairwise correlation of elicited confidence is $r = 0.749$ across four models, against $r = 0.646$ for correctness on the same items. Within model, confidence correlates with the answer stability of (6) across operating points ($r = 0.36$–0.50), and—critically—predicts equally well from another model's answer stability ($r = 0.30$–0.34), which rules out a within-episode circularity. Stability itself carries little information about correctness ($r = 0.05$–0.13). Confidence is therefore a readout of answer convergence on an item, and convergence is only weakly related to being right.

E. Discrimination is task-dependent, not a fixed model property

To test whether the account of Sections V-A and V-B predicts where confidence is usable, we evaluate the same models across conditions that differ in whether error arises from missing information or from missing knowledge. Fig. 4, Fig. 5 and Table IV report the primary endpoint across all conditions. Within a single model, AUROC ranges from 0.492 to 0.800. On standard-difficulty clinical items confidence is informative (0.699 and 0.703); on the hard subset of the same sources it is at chance (0.492 and 0.540). The pattern reproduces in a non-reasoning model (0.671 standard, 0.475 hard), so it does not depend on explicit reasoning machinery, and in a second vendor's model (0.480 on the hard subset).

Table XI confirms that the assignment is a resource manipulation and not merely a label: realised reasoning tokens move from a median of 688 to 0 across the ladder with no truncation anywhere. Table XII summarises the controls that address the alternative explanations of the results below, each of which is a run rather than an argument.

Two features of the frontier condition deserve note. First, the collapse is not a matter of degraded resolution: Fig. 5 shows the confidence distributions for correct and incorrect episodes to be essentially superimposed there, whereas at standard difficulty they separate visibly. Second, it is not confined to one vendor or scale. Across the four models evaluated on the hard split the interval for (2) contains 0.5 in three cases, and the one exception, gpt-5.4 at 0.540 $[0.517, 0.563]$, is the strongest model and still recovers less than a tenth of the discrimination available at standard difficulty. Reliability diagrams (Fig. 6) show the same failure from the calibration side: at the frontier, observed accuracy is flat at 0.28–0.42 across the entire range of stated confidence, so no threshold placed on that scale separates anything.

The magnitude of the swing bears emphasis. Within gpt-5.4-mini alone, (2) ranges from 0.492 to 0.751 across

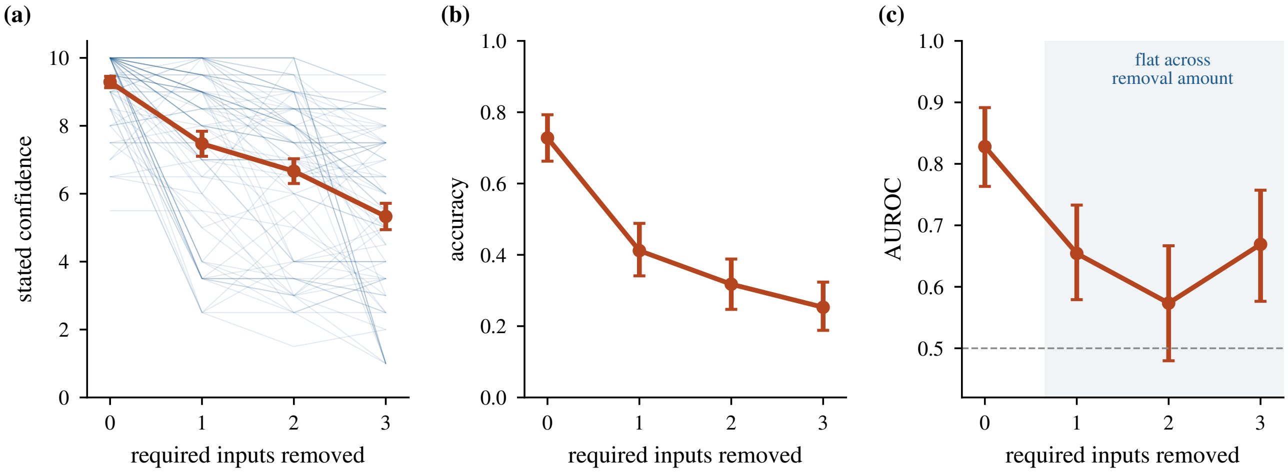

Fig. 3. Dose–response on MedCalc, gpt-5.4-mini, on the 85 items carrying at least three removable required inputs, so that every dose level is evaluated on the same items. (a) Stated confidence, with one faint line per item; the heavy line is the mean with a 95% interval. (b) Accuracy. (c) Discrimination, with intervals from an item-level bootstrap. Confidence and accuracy respond monotonically to how much of the requirement is removed; discrimination does not respond to the amount removed at all (shaded region), though it does fall at the first step away from a complete note.

conditions—a difference larger than that between the weakest and strongest models within any single condition. Model identity is therefore a poor predictor of confidence reliability relative to task condition, which is the practical content of the result.

TABLE IV
Discrimination of elicited confidence across conditions. AUROC with 95% bootstrap interval; ⋆ indicates the interval excludes 0.5.

Setting	Model	n	Acc.	AUROC [95% CI]
Clinical, static, capability frontier				
MedAgentsBench-hard	gpt-5.4-mini	2228	.387	.492 [.469, .515]
MedAgentsBench-hard	gpt-5.4	2235	.474	.540 [.517, .563]⋆
MedAgentsBench-hard	sonnet-5	2948	.445	.480 [.460, .499]
MedAgentsBench-hard	gpt-4o-mini	784	.157	.475 [.428, .523]
Clinical, static, standard difficulty				
MedAgentsBench-std	gpt-5.4-mini	1198	.760	.699 [.666, .729]⋆
MedAgentsBench-std	gpt-5.4	1200	.824	.703 [.666, .739]⋆
MedAgentsBench-std	gpt-4o-mini	800	.613	.671 [.637, .705]⋆
AgentClinic (static)	gpt-5.4-mini	409	.655	.669 [.617, .721]⋆
AgentClinic (static)	gpt-5.4	303	.667	.689 [.628, .748]⋆
Clinical, interactive consultation				
AgentClinic (inter.)	gpt-5.4-mini	642	.632	.751 [.713, .787]⋆
AgentClinic (inter.)	gpt-5.4	642	.727	.800 [.759, .840]⋆
Clinical, procedural				
MedCalc	gpt-5.4-mini	581	.690	.693 [.650, .736]⋆
Non-clinical controls				
MMLU-Pro	gpt-5.4-mini	600	.742	.704 [.659, .747]⋆
ARC-Challenge	gpt-5.4-mini	600	.958	.782 [.683, .874]⋆

F. Interactive consultation restores discrimination

On the same 214 AgentClinic cases, presented interactively rather than as complete vignettes, confidence discriminates at 0.751 [0.713, 0.787] and 0.800 [0.759, 0.840], against 0.669 and 0.689 for the static presentation of the same cases. Table II places the two presentations side by side. Deferral becomes useful: routing the least-confident 18.5% raises accuracy of the retained set from 0.632 to 0.715, and routing 37.5% raises it to 0.786 (Fig. 11c; Table VI). The stronger model shows the same pattern, from 0.727 to 0.859 at 24.8% deferral.

TABLE V
Per-source decomposition, gpt-5.4-mini. Under the standard split no source falls below chance; under the hard split four of ten do. The collapse is a property of the difficulty stratum rather than of any particular source.

	Capability frontier		Standard difficulty	
Source	Acc.	AUROC	Acc.	AUROC
AfriMedQA	.354	.319	.783	.518
MedExQA	.237	.410	.858	.588
PubMedQA	.153	.368	.672	.700
MedXpertQA-U	.310	.455	.375	.569
MMLU	.461	.504	.958	.841
MedXpertQA-R	.296	.504	.504	.557
MMLU-Pro	.400	.529	.817	.842
MedMCQA	.397	.550	.875	.802
MedBullets	.573	.596	.800	.720
MedQA	.671	.625	.950	.577

TABLE VI
Deferral value in interactive consultation, AgentClinic (214 cases × 3 seeds).

Model	Threshold	Deferred	Acc. retained	Δ
gpt-5.4-mini	none	0.0%	.632	—
	≥ 8	18.5%	.715	+.083
	≥ 9	37.5%	.786	+.153
	≥ 10	74.9%	.857	+.225
gpt-5.4	none	0.0%	.727	—
	≥ 8	24.8%	.859	+.132
	≥ 9	44.5%	.896	+.169
	≥ 10	71.5%	.907	+.180

G. The frontier collapse is not a base-rate artifact

To rule out class balance as the source of the frontier result, we repeat the comparison at matched accuracy. Because accuracy differs between the standard and hard splits, the AUROC difference could in principle reflect base

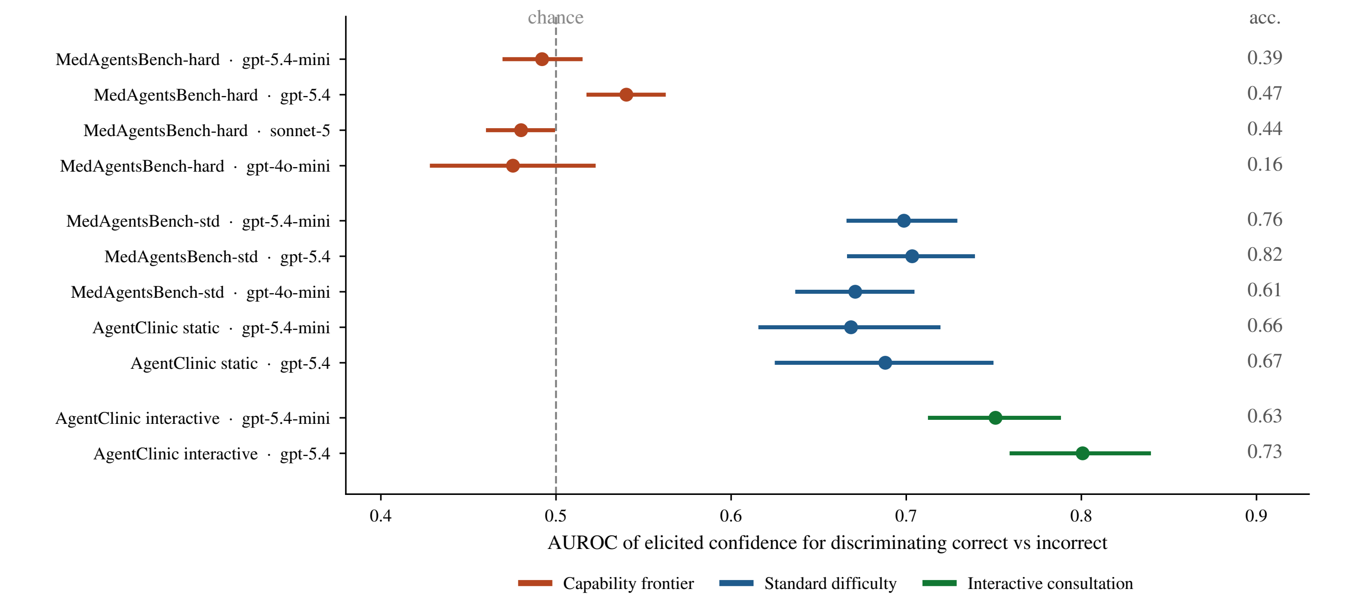

Fig. 4. Discriminative power of elicited confidence, by deployment condition. Points are AUROC with 95% bootstrap intervals over episodes; the right-hand column gives task accuracy. The same model spans chance to strong discrimination depending on condition: at the capability frontier, where the vignette is complete and the error is one of knowledge, confidence is uninformative; in interactive consultation, where the model must obtain its own evidence and may simply not have asked, it is most informative.

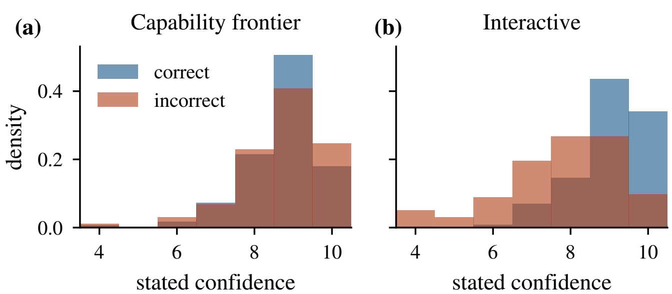

Fig. 5. Distribution of stated confidence for correct (blue) and incorrect (red) episodes, gpt-5.4-mini, at the two extremes of the range. (a) At the capability frontier the two distributions are superimposed. (b) Under interactive consultation they separate. The intermediate conditions interpolate between them.

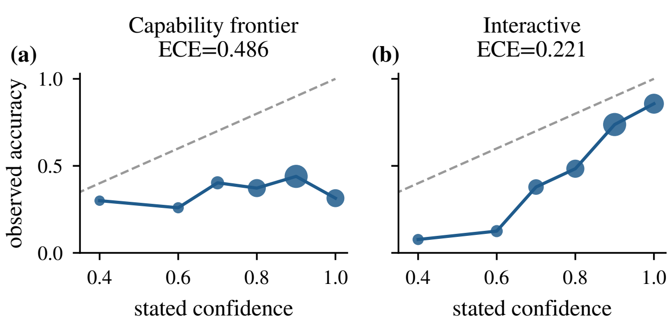

Fig. 6. Reliability diagrams, gpt-5.4-mini; marker area is proportional to bin occupancy and the dashed line is perfect calibration. (a) At the capability frontier the curve is essentially flat—observed accuracy is unrelated to stated confidence over its whole range. (b) Under interactive consultation it is monotone. Expected calibration error is reported above each panel; it moves in the same direction as discrimination without being equivalent to it.

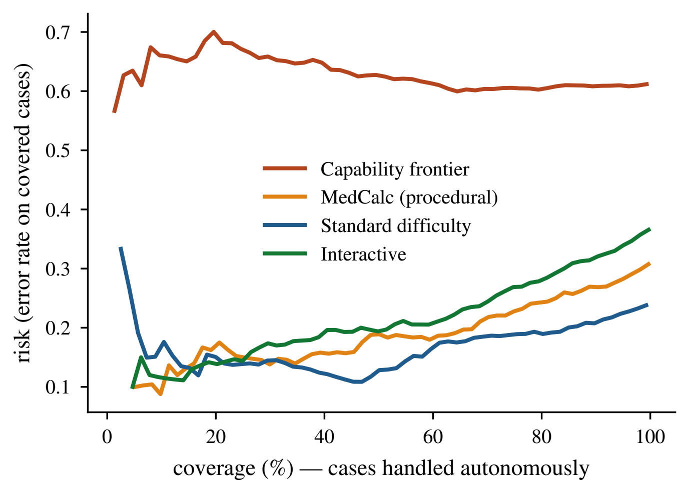

Fig. 7. Risk–coverage curves. Cases are handled autonomously in decreasing order of stated confidence. A useful confidence signal produces a curve rising steeply to the left; the capability-frontier curves are nearly flat, while the interactive-consultation curves rise steeply.

rate rather than difficulty. Applying the matching of (3) to the standard split leaves (2) unchanged: 0.698 [0.668, 0.731] at matched accuracy 0.387, and 0.703 [0.670, 0.735] at matched accuracy 0.474, against unmatched values of 0.698 and 0.703. The collapse is a property of the items, not of the class balance.

H. The failure is not a property of verbalization

Elicited confidence is one estimator among several. If its collapse at the frontier were an artifact of asking the model to put a number into words, an estimator that reads the model's own distribution should be immune. We therefore

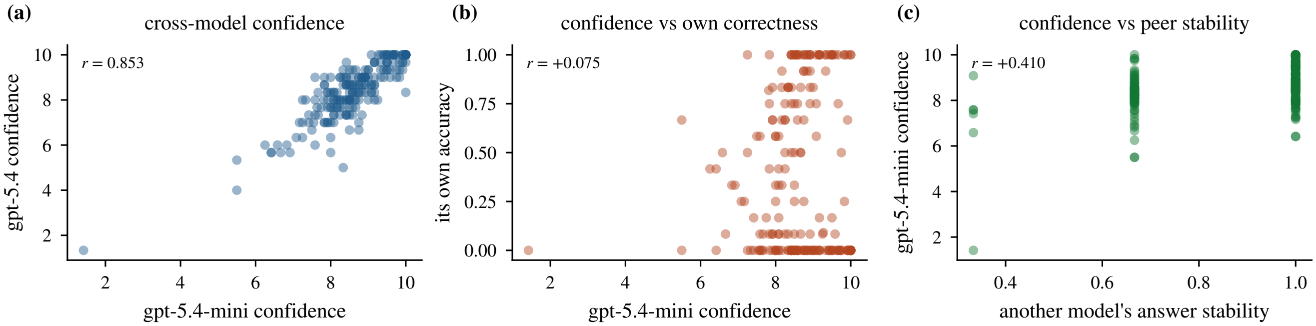

Fig. 8. What elicited confidence tracks. (a) Two models from different vendors agree strongly on which items warrant confidence. (b) Neither model's confidence predicts its own correctness. (c) A model's confidence is predicted about as well by another model's answer stability as by its own, which excludes within-episode circularity.

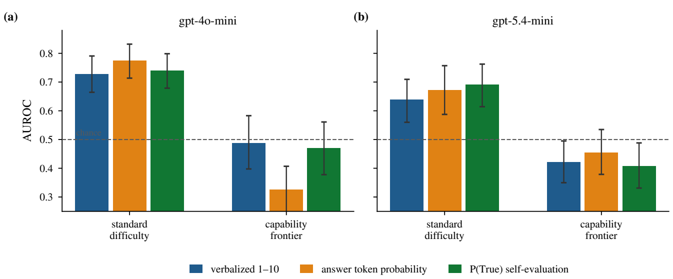

Fig. 9. Three confidence estimators measured on the same committed answers. At standard difficulty all three are informative; at the capability frontier all three sit at or below chance, and the token probability—which reads the model's own distribution rather than its words—is the worst of them on gpt-4o-mini. The failure is therefore not a failure of verbalization.

attach three estimators to the same committed answer, so that their AUROCs are scored against one correctness label and are directly comparable: the verbalized 1–10 value; the token probability the model assigned to the option letter it chose; and $P(\text{True})$, the probability it places on the token True when asked whether that answer is correct, which is the self-evaluation estimator of Kadavath et al. [2]. Two models expose token probabilities, one reasoning and one not, and each is measured on 250 standard and 250 frontier items.

All three behave the same way (Fig. 9, Table VII). At standard difficulty all are informative—0.729, 0.775, 0.740 on gpt-4o-mini and 0.638, 0.673, 0.691 on gpt-5.4-mini. At the frontier all six values fall at or below chance, between 0.325 and 0.488. The paired contrast against the verbalized value gives no alternative an advantage there: $P(\text{True})$ is indistinguishable from it on both models ($-.017$ $[-.088, +.054]$ and $-.013$ $[-.088, +.059]$), and the token probability is significantly worse on gpt-4o-mini ($-.163$ $[-.276, -.045]$), where it reaches 0.325—an inversion strong enough that ranking by it is worse than ranking at random.

This is the control that decides how far the paper's claim reaches. Verbalization is not the bottleneck, and neither is the elicitation format (Section V-I) nor the response scale (Section V-J): the constraint is on what the model has to report. Where the vignette is complete and the residual error is one of medical knowledge, no read-out of the model—in words, in logits, or in a second self-evaluating pass—separates its right answers from its wrong ones.

TABLE VII
Three estimators on identical committed answers, AUROC with 95% bootstrap interval, $n = 250$ per cell. Δ is the paired contrast against the verbalized value at the frontier.

Model / stratum	Verbalized 1–10	Token probability	$P(\text{True})$
gpt-4o-mini, standard	.729 [.666,.790]	.775 [.716,.833]	.740 [.679,.801]
gpt-4o-mini, frontier	.488 [.395,.573]	.325 [.249,.404]	.470 [.380,.561]
Δ vs. verbalized	—	$-.163$ $[-.276, -.045]$	$-.017$ $[-.088, +.054]$
gpt-5.4-mini, standard	.638 [.565,.713]	.673 [.588,.758]	.691 [.619,.763]
gpt-5.4-mini, frontier	.422 [.354,.495]	.456 [.375,.534]	.408 [.329,.488]
Δ vs. verbalized	—	$+.034$ $[-.049, +.116]$	$-.013$ $[-.088, +.059]$

I. No elicitation format recovers discrimination

To test whether the null reflects how confidence was requested rather than what the model has to report, we compare four elicitation formats and one non-verbal alternative on identical items. Across four formats and two operating points, all eight cells lie at chance: integer 0.509/0.451, probability 0.540/0.451, verbal 0.486/0.490, post-hoc self-check 0.522/0.443. Repeated sampling is likewise uninformative: within a fixed operating point, self-consistency across three samples discriminates no better than the verbalized value (0.500–0.562 versus 0.527–0.563). The failure is therefore not one of verbalization—there is no accessible signal being poorly reported.

J. Sensitivity to elicitation design

To confirm that no aspect of the response schema is load-bearing, we ablate its three degrees of freedom on identical items. Three aspects of the elicitation schema could in principle drive the results: the cap on the elicited differential, the position of the confidence field relative to the answer, and the granularity of the confidence scale. Table VIII reports each.

Varying the differential cap $K \in \{3, 5, 10\}$ moves elicited breadth as intended ($2.91 \rightarrow 3.82 \rightarrow 4.15$ at standard difficulty) while leaving discrimination unchanged (0.690, 0.682, 0.724; all intervals overlapping), and the frontier condition is likewise unmoved (0.490, 0.498, 0.507). The schema's ceiling therefore governs how much the model lists, not how well its confidence separates correct from incorrect answers—which also means the breadth measure and the discrimination measure are not two views of one quantity.

Moving the confidence field ahead of the answer, so that the model commits to a confidence before committing to a response, changes nothing at standard difficulty (0.692 versus 0.682) and if anything weakens the frontier condition further (0.447 versus 0.496, intervals overlapping). Joint generation of answer and confidence is therefore not what produces the null: a confidence stated before the answer exists is no more informative than one stated after.

Coarsening the response scale from ten levels to five costs discrimination at standard difficulty (0.764 → 0.642) without changing the frontier condition (0.477 versus 0.499, overlapping). The scale therefore limits how finely a usable signal can be expressed but does not create one where none exists—which is the expected pattern if the constraint is on the information available rather than on the channel carrying it.

TABLE VIII
Sensitivity to elicitation design, gpt-5.4-mini, 150 items per split. Varying the differential cap changes elicited breadth but not discrimination; presenting the confidence field before the answer does not change it either. Neither aspect of the schema is load-bearing.

Ablation	Variant	Acc.	Breadth	AUROC	[95% CI]
Standard difficulty					
Differential cap K	$K = 3$	.767	2.91	.690	[.624, .751]
	$K = 5$ (default)	.766	3.82	.682	[.617, .745]
	$K = 10$	.747	4.15	.724	[.662, .785]
Confidence position	after answer (default)	.766	3.82	.682	[.615, .745]
	before answer	.755	3.78	.692	[.627, .754]
Confidence scale	1–5	.750	—	.642	[.573, .709]
	1–10 (default)	.773	—	.764	[.703, .820]
Capability frontier					
Differential cap K	$K = 3$	.339	2.91	.490	[.427, .553]
	$K = 5$ (default)	.353	3.88	.498	[.437, .561]
	$K = 10$	.367	4.31	.507	[.446, .570]
Confidence position	after answer (default)	.353	3.88	.496	[.435, .559]
	before answer	.380	3.84	.447	[.385, .510]
Confidence scale	1–5	.347	—	.499	[.437, .560]
	1–10 (default)	.363	—	.477	[.415, .539]

TABLE IX
Cost of transferring a deferral threshold across conditions. The threshold is the smallest value achieving ≥ 0.85 accuracy on the retained set in the calibration condition; it is then applied unchanged in the deployment condition. Bold marks a deployment accuracy below the service level the threshold was chosen to guarantee; any marks a calibration condition in which no threshold attains it.

Model	Calibrated on	Deployed on	τ	Deferred	Acc. retained
gpt-5.4-mini	standard bench	static OSCE	10	84.6%	.857
	standard bench	interactive	10	74.9%	.857
	standard bench	frontier	10	80.4%	.352
	static OSCE	frontier	10	80.4%	.352
	interactive	frontier	10	80.4%	.352
	frontier	any	—	—	no τ attains .85
gpt-5.4	standard bench	interactive	9	44.5%	.896
	standard bench	frontier	9	45.3%	.497
	interactive	frontier	8	22.2%	.486
	frontier	any	—	—	no τ attains .85

TABLE X
The central experiment: removing and restoring the single item a case requires, across three models spanning two orders of magnitude of scale and two clinical task types. In AgentClinic the removed item is the key diagnostic finding, which leaves the case harder but answerable; in MedCalc it is a numerically required laboratory value, without which the calculation cannot be performed. Stated confidence tracks the severity of the deficiency—3.5 points where the case becomes unanswerable, 0.5 where it remains answerable—while discrimination does not move in three of four cells and is marginal in the fourth.

Task	Model	Acc.	Conf.	AUROC	[95% CI]
MedCalc: required input removed (case becomes unanswerable)					
	gpt-5.4-mini	.302	5.72	.728	[.681, .772]
		.712	9.30	.791	[.747, .832]
	Δ restored	+.411 /	+3.58	+.064	[+.001, +.130]
	gpt-5.4	.344	5.59	.760	[.718, .802]
		.752	9.08	.788	[.742, .832]
	Δ restored	+.408 /	+3.49	+.028	[−.035, +.089]
AgentClinic: key finding removed (case remains answerable)					
	gpt-5.4-mini	.764	8.96	.726	[.669, .781]
		.855	9.43	.732	[.661, .800]
	Δ restored	+.091 /	+0.46	+.006	[−.085, +.095]
	gpt-5.4	.811	8.81	.755	[.690, .817]
		.897	9.36	.692	[.609, .774]
	Δ restored	+.086 /	+0.55	−.061	[−.167, +.042]
	gpt-5.4-nano	.729	7.09	.738	[.686, .787]
		.834	7.77	.792	[.739, .845]
	Δ restored	+.105 /	+0.68	+.054	[−.022, +.128]
Evidence agency, AgentClinic (acquired on request vs. given)					
	gpt-5.4-mini			+.039	[−.039, +.118]
	gpt-5.4			+**.107**	[+.008, +.204]
	gpt-5.4-nano			+.049	[−.015, +.114]

K. The free-text scoring rule does not drive the result

Free-text diagnoses are scored by the alias-aware match of (4), which is strict: it marks "Pancoast tumor" wrong against a reference of "apical lung tumor", and "Jersey finger" wrong against "rupture of the flexor digitorum profundus tendon". Because the mis-scored answers tend to be confident ones, the rule could in principle depress measured discrimination. We therefore re-adjudicated all 564 distinct (reference, candidate) pairs that the string rule marked wrong, using a judge model that is not among the evaluated models and is asked only whether the two strings name the same clinical entity; 10.1% were judged equivalent. Every principal quantity was then recomputed under the relaxed scorer.

Nothing that the paper concludes depends on the choice. Static AgentClinic is unchanged to three decimals in both accuracy and (2). Interactive consultation improves under the relaxed rule—accuracy 0.632 → 0.659 and 0.727 → 0.757, discrimination 0.751 → 0.773 and 0.800 → 0.837—so the condition this paper reports as most favourable to confidence gating is, if anything, understated by the strict rule. In the removal manipulation the accuracy contrast grows slightly (+.091 → +.103, +.086 → +.100, +.105 → +.107) and the discrimination contrast remains within the interval containing zero in two of the three cells; in the third, gpt-5.4-nano, the relaxed interval excludes zero at +.086 [+.015, +.157], which we report rather than resolve. Strict scoring is used throughout the paper because it is the pre-registered rule; the relaxed values are given here so that the dependence can be checked.

L. The pattern holds on independent corpora

To establish that the pattern is not an artifact of one benchmark family, we repeat the measurement on corpora constructed separately from the clinical sets. Table V decomposes the two clinical strata by source. Under the standard split no source falls below chance. Under the hard split four of ten do—AfriMedQA at 0.319, PubMedQA at 0.368, MedExQA at 0.410, MedXpertQA-U at 0.455—so on those sources stated confidence is not merely uninformative but inverted: the answers the model is more confident about are, if anything, more often wrong. The two sources retaining the most discrimination at the frontier, MedQA and MedBullets, are also the two on which the model is most accurate there, consistent with the association reported above.

The clinical sets share a benchmark family, so the standard-versus-frontier contrast could reflect a property of that construction. Three independent corpora reproduce the association between competence and discrimination: MMLU-Pro at accuracy 0.742 gives 0.704 $[0.659, 0.747]$, ARC-Challenge at 0.958 gives 0.782 $[0.683, 0.874]$, and the free-response MedCalc set at 0.690 gives 0.693 $[0.650, 0.736]$. Answer format is not the operative variable: MedCalc is numeric free-response and behaves like the multiple-choice sets. Fig. 10 places all seven sets on common axes. The association between competence and discrimination is visible, and so is the systematic departure from it that this paper is about: interactive consultation sits well above the trend, because there the model's errors are errors of insufficient evidence, which is what its confidence reports.

TABLE XI
First stage: realised reasoning tokens by assigned operating point, gpt-5.4-mini. The assignment moves the resource by more than an order of magnitude with no truncation, which is what licenses treating the nominal effort setting as a resource manipulation rather than a label.

Operating point	n	Median	IQR	p90	Trunc.
high	745	688	[284, 1846]	3375	.000
low	745	80	[42, 159]	317	.000
none	745	0	[0, 0]	0	.000

M. Significance and equivalence of the principal contrasts

Two of this paper's three claims assert a difference and one asserts its absence, and the two need different tests. Differences are tested by Wilcoxon signed-rank on item-paired values where the design is paired and by a 4000-permutation test on the condition label where it is not; Holm correction is applied across the family of nine difference tests. An interval containing zero is not evidence of no effect, so the null claim is tested separately, by two one-sided tests against a pre-specified equivalence margin of 0.10 AUROC—about a third of the range this paper

TABLE XII
Threats to validity and the design element that addresses each. Every row is a control run rather than an argument; the section giving the result is named.

Threat	Control	Result
Class balance	accuracy-matched resampling (3)	V-F: unchanged
Manipulation is binary	graded removal of 1, 2, 3 inputs	V-B: AUROC flat
Response format	four elicitation formats	V-G: all at chance
Response scale	1–5 vs 1–10 vs 1–100	V-H: no signal created
Elicitation position	confidence before vs after answer	V-H: unchanged
Differential cap	$K \in \{3, 5, 10\}$	V-H: breadth only
Poor verbalisation	sampling self-consistency	V-G: no better
Ceiling censoring	escalation rule, per-episode log	IV-A: zero truncation
Corpus idiosyncrasy	three independent corpora	V-I: reproduces
Vendor	two vendors	V-D: reproduces
Architecture	non-reasoning model	V-D: reproduces

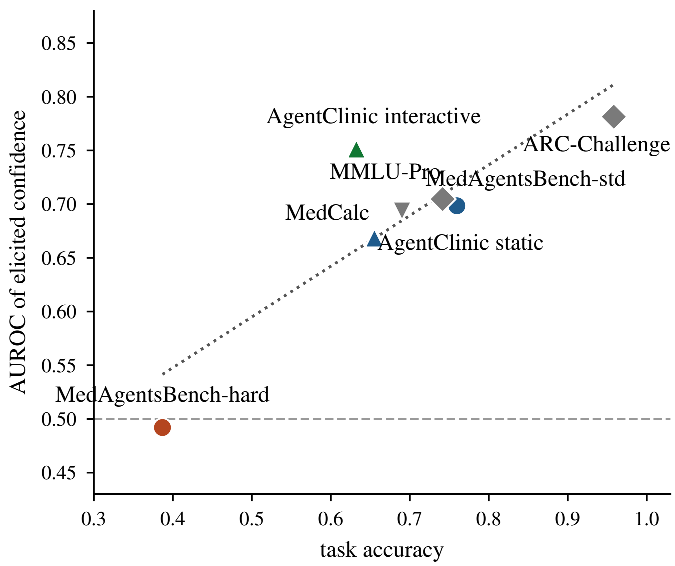

Fig. 10. Discrimination against task accuracy across all seven evaluation sets, gpt-5.4-mini. Competence and discrimination are associated (dotted trend), but competence does not explain everything: interactive consultation sits well above the trend, and it is the one condition in which the model must obtain its own evidence.

reports across deployment conditions, and below any shift that would change a deferral policy. The equivalence tests form their own Holm family.

Every difference claim survives correction with room to spare (Table XIII). The manipulation moves confidence at $p_{\text{Holm}} \leq 7 \times 10^{-25}$ in all five cells and accuracy at $p_{\text{Holm}} \leq 3 \times 10^{-6}$; the frontier-versus-standard and static-versus-interactive contrasts survive at $p_{\text{Holm}} \leq 0.021$; the dose trend in confidence is at 1×10^{-61}.

The equivalence tests are the honest limit of the null claim. Formal equivalence at the 0.10 margin is attained in one of the five cells (MedCalc / gpt-5.4, $p_{\text{Holm}} = 0.050$) and not in the others, whose p_{Holm} values run from 0.074 to 0.32. The design is therefore powered to show that the discrimination contrast is small, not to prove it is exactly zero. What the data support is a bounded statement: in no cell does the contrast differ significantly from zero, and in every cell the 95% interval excludes a shift larger than 0.17 in either direction. Set against a manipulation that moves accuracy by 41 points and confidence by 3.6, that bound is the result. We state the claim in those terms

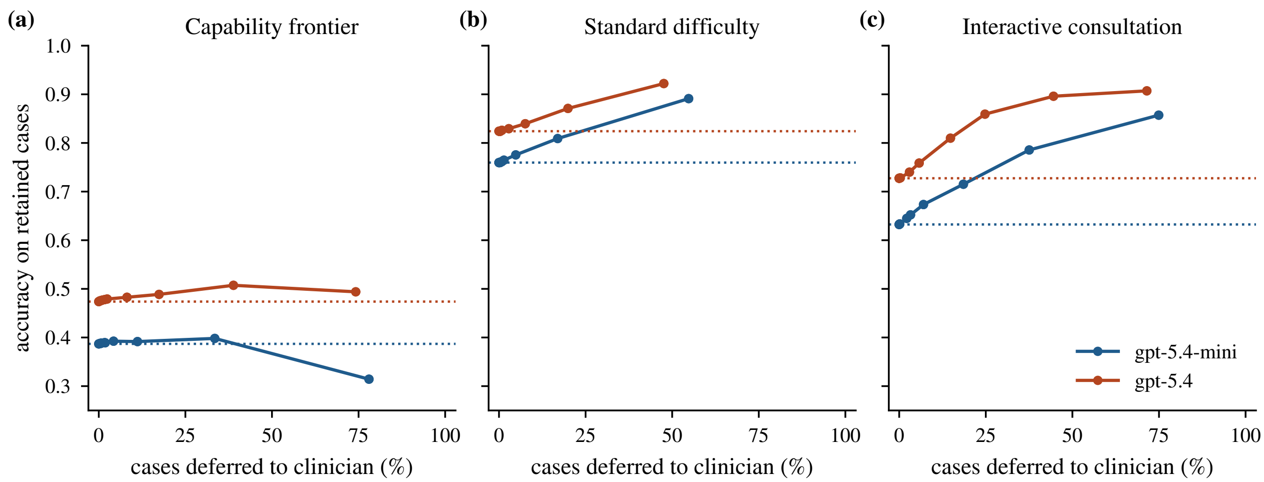

Fig. 11. Decision curves for confidence-gated deferral, endpoint diagnostic accuracy, evaluated by (5). Cases below the threshold are routed to a clinician; dotted lines mark the no-deferral baseline. At the capability frontier (a) deferring cases buys almost nothing and for gpt-5.4-mini turns harmful past 60%; on complete standard vignettes (b) it buys a moderate amount; in interactive consultation (c), where the model obtains its own evidence, the same mechanism on the same case material yields large monotone gains. The gate's value is set by the condition, not by the model.

throughout rather than as a demonstrated null.

TABLE XIII
Tests of the principal contrasts. Difference claims and equivalence claims are corrected as separate Holm families. The equivalence margin is 0.10 AUROC, fixed before testing.

Claim	Cell	Estimate	p_{Holm}
Difference claims (Wilcoxon signed-rank / permutation)			
Removal moves confidence	MedCalc, both models	+3.49, +3.59	$\leq$ 7e–57
Removal moves confidence	AgentClinic, 3 models	+0.47 to +0.68	$\leq$ 7e–25
Removal moves accuracy	MedCalc, both models	+.408, +.411	$\leq$ 3e–34
Removal moves accuracy	AgentClinic, 3 models	+.086 to +.105	$\leq$ 3e–06
Standard > frontier AUROC	gpt-5.4-mini / gpt-5.4	+.207 / +.163	1.3e–03
Interactive > static AUROC	gpt-5.4-mini / gpt-5.4	+.083 / +.111	.021 / .009
Dose lowers confidence	gpt-5.4-mini	$\rho = -.582$	1.4e–61
Dose 1 vs 3 moves AUROC	gpt-5.4-mini	+.015	.81 (n.s.)
Equivalence claims (TOST, margin 0.10)			
Removal does not move AUROC	MedCalc / gpt-5.4	+.029	**.050**
Removal does not move AUROC	AgentClinic / gpt-5.4-mini	+.006	.074
Removal does not move AUROC	MedCalc / gpt-5.4-mini	+.062	.32
Removal does not move AUROC	AgentClinic / gpt-5.4	−.059	.32
Removal does not move AUROC	AgentClinic / gpt-5.4-nano	+.054	.32

N. The cost of transferring a threshold

To convert the preceding results into a deployment quantity, we calibrate a deferral threshold in one condition and apply it unchanged in another. The recommendation that a gate be validated where it will run is only useful if the cost of ignoring it can be stated. We therefore calibrate a threshold in one condition and apply it unchanged in another. Fixing a service level of 0.85 accuracy on the retained set, Table IX gives the smallest threshold meeting it in the calibration condition and the accuracy that threshold actually delivers elsewhere.

Transfer into the capability frontier fails from every origin tested. A threshold calibrated on standard benchmark items, on static OSCE vignettes, or in interactive consultation delivers 0.352 and 0.486–0.497 at the frontier—between 35 and 50 points below the service level it was chosen to guarantee—while still deferring 22% to 80% of cases. The gate therefore costs the full clinician workload it was budgeted for and returns less than half the accuracy it promised. The shortfall is invisible from the calibration condition: the same threshold meets its target there, so nothing in the validation data signals that it will not hold.

Transfer in the reverse direction is benign. A threshold calibrated on standard items exceeds its target when applied to interactive consultation (0.857 and 0.896), because discrimination is better in that setting than where the threshold was set. The asymmetry matters operationally: the direction of transfer that a development pipeline naturally takes—validate on a benchmark, deploy to a harder stratum—is the direction that fails.

Calibrating at the frontier is not a remedy. For neither model does any threshold reach 0.85 there: the calibration procedure returns a degenerate setting that defers every case. A gate calibrated where the model is not competent does not produce a conservative operating point, it produces the abandonment of autonomy—a different and more honest failure than silently under-performing, but not a deployable one.

This is the account of Section V-A stated in units a deployment can act on. Discrimination is high where the model's errors are errors of missing evidence, and a threshold set there encodes a relationship between confidence and correctness that does not exist at the frontier, where the evidence is complete and the error is one of knowledge.

VI. Discussion

The practical conclusion is that "is elicited confidence reliable?" has no model-level answer. The same model, on clinically similar material, spans chance to strong discrimination depending on whether the item is at its capability frontier and whether the model gathered the evidence itself. Reporting a single calibration figure for a clinical LLM is therefore not meaningful for deployment.

Two consequences follow for system design. First, confidence-gated deferral cannot be validated on standard benchmark items and then deployed at the capability frontier: on exactly the cases where a safety mechanism is most needed, discrimination of answer correctness drops from 0.70 to 0.48–0.54, and a threshold carried across delivers 0.35–0.50 against an 0.85 target. Second, and in the opposite direction, static benchmark evaluation understates confidence reliability for consultation agents. The two results are not in tension: a consultation agent's errors are largely errors of evidence it never gathered, which is exactly what its stated confidence reports, whereas a fully specified vignette never requires the model to earn its evidence and leaves only knowledge errors, about which the signal is silent. Section VI-A develops this account and reports the test that rules out the alternative.

A. What the signal is, and why the conditions differ

The two headline observations—confidence is uninformative at the frontier yet strong in interactive consultation—are reconciled by what the signal reports. The natural explanation, that acquiring the discriminating evidence makes convergence and correctness share a cause, is directly testable and fails: supplying that evidence changes accuracy and stated confidence without changing discrimination at all (Section V-A).

What survives the manipulation is an account in terms of information state. Elicited confidence tracks, imperfectly, whether the model has what it needs to answer—it separates episodes in which the key finding was obtained from those in which it was not at AUC = 0.661—and it does not track whether the answer drawn from that information is right. Discrimination of correctness is therefore high exactly where insufficiency is the dominant error mode and low where it is not. In interactive consultation the model frequently fails to ask, so insufficiency dominates and confidence is informative. At the capability frontier every item is fully specified; errors are failures of medical knowledge rather than of information, and a signal about information carries nothing about them. The subgroup inversion follows for the same reason: among episodes where the model did not obtain the key finding, discrimination is higher (0.790) than where it did (0.714), because in the former group the errors are the kind the signal can see.

This also explains why no elicitation format helps. All four formats read out the same underlying quantity; rewording the question cannot make a signal about information sufficiency report on knowledge adequacy. It is consistent with general-domain findings that cross-model confidence is dominated by a shared item factor [14] and that overconfidence arises from insufficient conditioning on the model's own answer [6]—an answer-conditioned signal is exactly what an information-state signal is not.

B. Implications for clinical deployment

Three consequences follow for systems that gate on confidence.

A gate must be validated in the modality it will run in. Confidence measured on static question sets underestimates its reliability for a consultation agent by roughly 0.08 AUROC in our data, and confidence measured in consultation would overstate reliability for a system that answers from a completed record. Neither direction of extrapolation is safe, and the difference is larger than the difference between the models we tested.

A gate must be validated on the difficulty stratum it will run on, and the cost of not doing so is measurable rather than hypothetical. Table IX shows a threshold meeting an 0.85 service level in validation returning 0.35–0.50 at the frontier, from every origin tested, with nothing in the validation data to signal it. The failure is directional: gates transfer safely from complete vignettes into consultation, where the model earns more of its own evidence, and unsafely into the frontier, where evidence is not what it lacks.

Deferral must be assessed on a decision curve, not a calibration statistic. A threshold chosen to optimise ECE cannot express any of this, because the failure is in which cases are ranked highest, not in the scale on which they are ranked. Fig. 11 makes the difference visible: the same mechanism on the same model yields steep monotone gains in consultation and a flat curve at the frontier, and no summary of the confidence scale distinguishes those two situations.

C. Relation to prior findings

Where this paper overlaps the general-domain literature it agrees with it, and we treat the agreement as a validity check on the instrument rather than as a finding: cross-model consistency in stated confidence, near-zero within-model correlation with correctness, and no recovery from alternative elicitation all reproduce what has been reported for non-clinical benchmarks [14], [3]. The clinical results have no counterpart in that literature because it does not manipulate evidence: the observation that removing a required finding moves stated confidence without moving discrimination requires a task in which one can identify what a case needs, remove exactly that, and score the answer—which diagnostic vignettes and clinical calculations supply and general-knowledge benchmarks do not.

D. Limitations

The evidence manipulation identifies the required item automatically, from a pre-registered annotation of each case's key discriminating finding; no physician reviewed those annotations, and on AgentClinic the removal is verified to change the answer only through a mechanical check that the withheld string no longer appears in the vignette. The manipulation removes one item, so it bounds what confidence reports about a single deficiency rather than about degrees of incompleteness. Both vendors' frontier models were evaluated, but the third vendor's arm is limited to the capability frontier. Confidence was elicited jointly with the answer in a single response for the principal conditions, which is the deployment-realistic protocol but confounds elicitation with generation; the post-hoc format controls for this and shows no difference.

VII. Conclusion

Elicited confidence in clinical language models is not a stable property that can be measured once and relied upon. Across 34,200 episodes, five models, two vendors and seven evaluation sets, its discriminative power ranges from chance to strong within a single model, governed by whether the item lies at the model's capability frontier and whether the model gathered the evidence itself. The variation across conditions exceeds the variation across models.

The reason is not that confidence is noise. It is a reading of whether the model has what the case requires: removing the one item a case needs moves stated confidence by up to 3.6 points on a ten-point scale while shifting discrimination by no more than .17 AUROC, undetectably, in every cell tested. That single fact predicts both extremes. Where errors arise from evidence the model never obtained, its confidence sees them, and deferral in interactive consultation reaches 0.75–0.80 and is genuinely useful. Where the evidence is complete and the error is one of knowledge, there is nothing for the signal to report, and discrimination sits at chance.

The cost of ignoring this is quantified rather than asserted: a threshold meeting an 0.85 service level in validation returns 0.35–0.50 at the capability frontier while still deferring up to four fifths of cases, and the development pipeline's natural direction of transfer—validate where the model is competent, deploy where it is not—is the failing one.

Confidence-gated deferral is therefore neither uniformly sound nor uniformly unsound, and which it is can be predicted in advance. It is sound where the model's remaining errors are errors of missing information, and unsound where they are errors of knowledge. That distinction, not a calibration statistic, is what a clinical deployment needs to establish before it gates on confidence.

Data and Code Availability

All benchmarks are publicly available. Episode-level records, the key-finding annotations and manipulation scripts underlying Section V-A, and all analysis code are released with the paper.

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